Empowering Open Source with Azure: From Code to Cloud

This session covers theory, real-world examples, and a live demo.



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Demystifying the Cloud: From Google Drive to AWS



Computing Over the Internet

Access resources and services remotely.



Daily Life Examples

Google Drive, YouTube, and Gmail are all cloud-powered.



Professional Platforms

Transition to AWS, Azure, and Google Cloud Platform.

Cloud computing means using shared computing resources—servers, storage, databases, networking, software, analytics, and intelligence—over the Internet ("the cloud") instead of owning them directly.

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Real-World Cloud Use Cases

Netflix: Global Streaming

Netflix leverages AWS to deliver content worldwide.

Ensures seamless streaming for millions of users.

Startups: Rapid Scaling

New companies use Google Cloud to scale websites quickly.

Launch and expand in minutes, not months.

COVID-19 data dashboards were hosted on Microsoft Azure, providing critical insights.



Healthcare IoT for real-time patient monitoring.



Real-time analytics for instant business insights.



Automated customer support via AI chatbots.

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The Role of Cloud in Modern Business



Digital Backbone

Cloud serves as the foundation for digital transformation across all industries.



Edge Computing Growth

The edge computing market is projected to reach $274B by 2025, expanding cloud reach.



Cloud-Native Workloads

By 2025, 95% of new digital workloads will be built cloud-native, enhancing efficiency.

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AI-Driven Cloud Services

Embedded AI/ML

Cloud platforms now seamlessly integrate AI and machine learning.

Automated Operations

AI automates complex operations and refines decision-making processes.

Insight Extraction

It efficiently extracts valuable insights from vast, complex datasets.



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Cloud-Native Architecture: The New Standard

Agility with Microservices

Microservices, containers, and Kubernetes drive unparalleled agility.

New Application Focus

95% of new applications will run on cloud-native platforms by 2025.

Resilience and Scalability

This architecture enables resilience, scalability, and faster market entry.

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Cloud Security and Compliance Trends



Enhanced Protection

New tools provide stronger defenses against evolving cyber threats.

Zero-Trust Models

Zero-trust, encrypted traffic, and automated compliance are standard.

Top Investment

Security remains a primary investment area in cloud infrastructure.

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Serverless and Event-Driven Architectures



Serverless Benefits

Reduces operational overhead and adopts a flexible pay-per-use billing model.

Rapid Development

Supports quick development and scalable responses for unpredictable workloads.

Event-Driven

Allows automatic responses to real-time business events and triggers.

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Edge Computing: Processing Data at the Source

Reduced Latency

Data processing occurs near users and devices, significantly cutting latency.

Key for Real-time

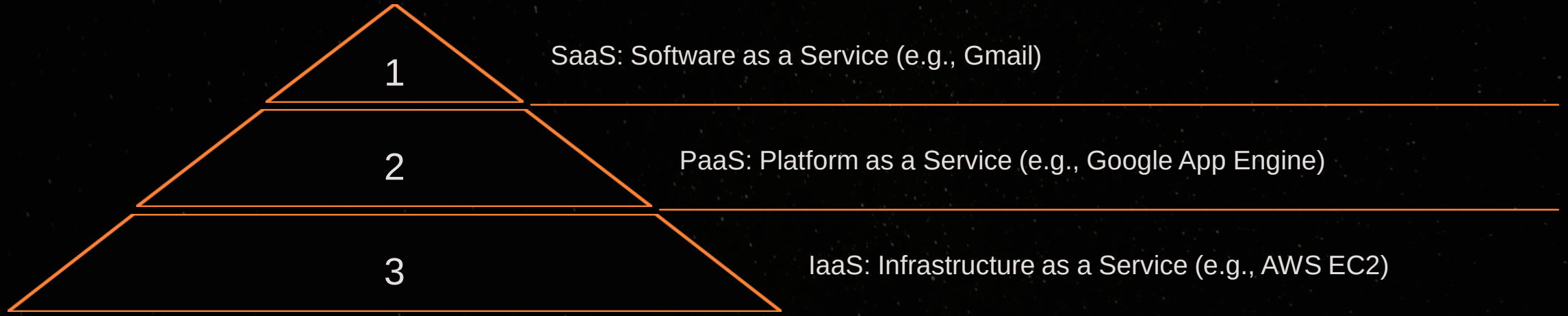
Crucial for IoT, autonomous vehicles, and real-time applications.

Enterprise Data Shift

By 2026, 75% of enterprise data will be processed at the edge, decentralizing operations.

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Cloud Service Models: IaaS, PaaS, SaaS



Understand the levels of cloud abstraction and control.

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Hybrid and Multi-Cloud Approaches

Flexibility in Deployment

Blends public clouds, private clouds, and on-premises infrastructure for optimal choice.

Addressing Core Needs

Manages data residency, optimizes costs, and enhances performance efficiently.

Widespread Adoption

A significant 87% of organizations are adopting multi-cloud strategies for resilience and scale.

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Cloud Deployment Types

Public Cloud

Shared resources owned by a third-party provider (e.g., AWS, Azure).

Offers high scalability and cost-effectiveness.

Private Cloud

Dedicated to one organization for exclusive use.

Provides greater control, security, and customization.

Hybrid Cloud

Combines public and private clouds, connecting resources.

Achieves flexibility with sensitive data control.



Cloud Storage Wars

- **Google Drive:** Personal file storage and collaboration.
- **Dropbox:** Simple, cross-device file sync and sharing.
- **AWS S3:** Scalable object storage for developers and businesses.

Object storage (like S3) treats data as objects in buckets. This differs from file or block storage.



You can easily upload and download files, just like using a folder.

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Live Demo: Launch Your First Virtual Machine

Choose Your OS

Select your preferred operating system.

Configure Resources

Set CPU, memory, and storage specifications.

Select Region

Choose the nearest geographic server location.

Launch and Connect

Start your VM and establish a connection.

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Virtual Machine vs. Container: What's the Difference?

Virtual Machines (VMs)

VMs include a full operating system.

They offer complete isolation but have higher overhead.

Containers (e.g., Docker)

Containers are lightweight and share the host OS kernel.

They provide efficient resource utilization.

Think of VMs as separate houses and containers as apartments within one building.

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APIs in the Cloud: Why and How?

Programmatic Control

APIs enable automation and management of cloud resources.

Command Line Example

Start a VM using a simple CLI command.

Python SDK Integration

Utilize Python SDKs for complex cloud operations.

We'll do a quick hands-on demonstration of a simple cloud API call.



APIs That Power the Internet

What is an API?

It's an Application Programming Interface. It lets different software talk to each other.

Weather Data

Apps use weather APIs to display forecasts.

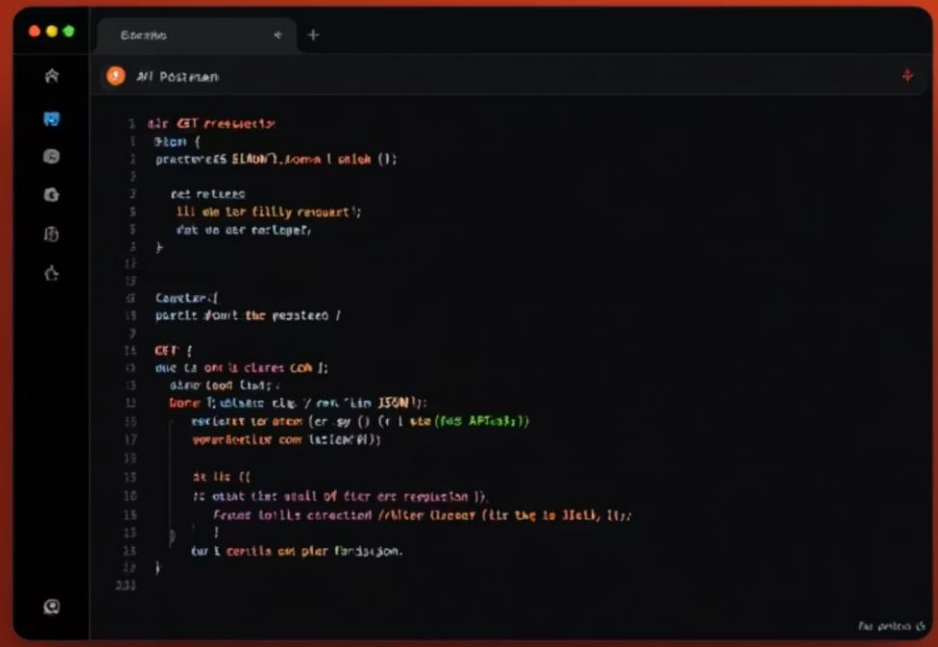
Translation Services

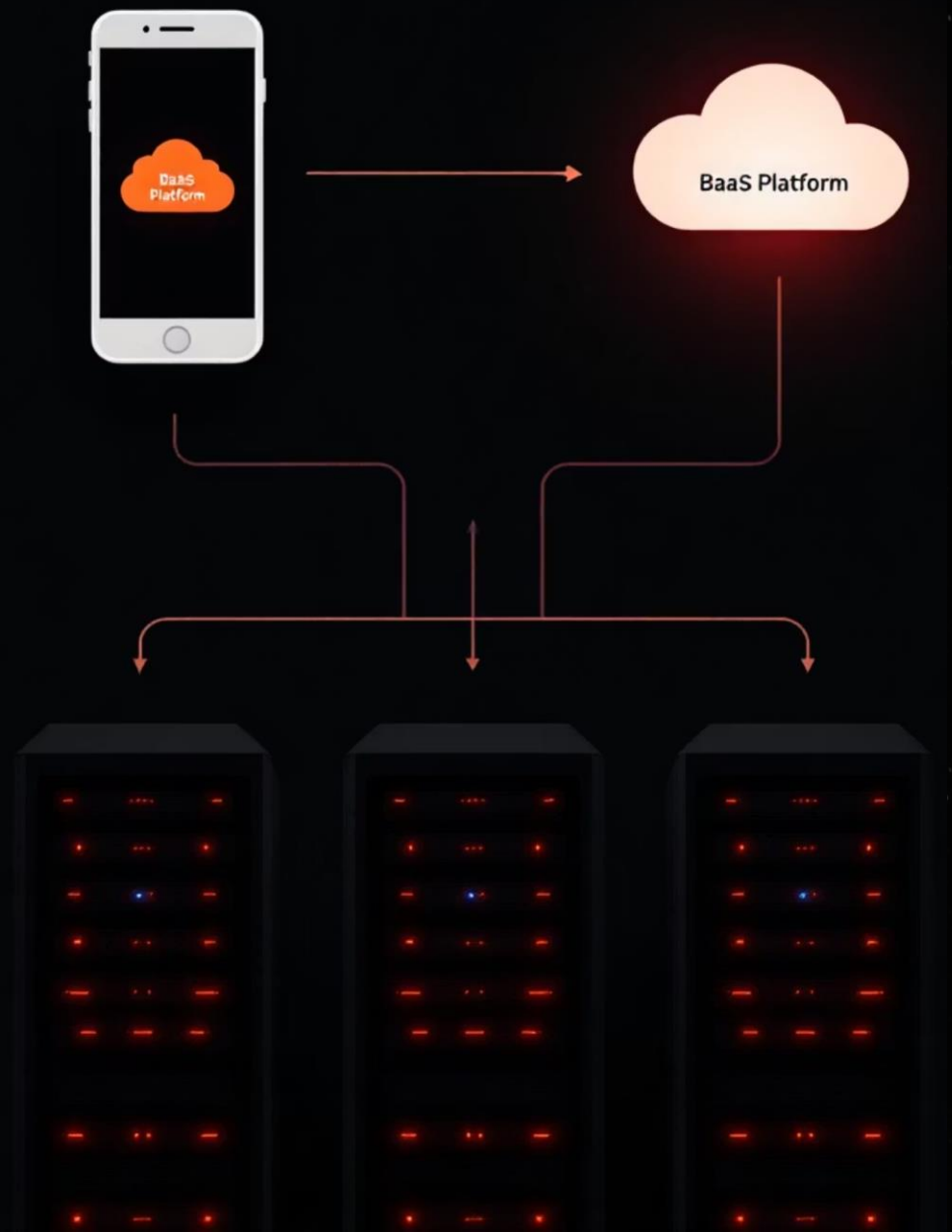
Google Translate uses APIs for language conversion.

Map Integration

Maps APIs power navigation and location services.

APIs enable seamless integration and data exchange across countless applications, forming the backbone of modern web services. They define the methods of communication between various software components.





Backend-as-a-Service (BaaS)

Platforms

Firebase, Supabase, and AWS Amplify simplify backend development.

Use Case

Build and host apps quickly. No server setup required.

Simplified Development

Focus on your app's frontend. BaaS handles server logic and scaling.

BaaS abstracts away the complexities of server-side infrastructure. It provides pre-built services for user authentication, databases, storage, and more. This accelerates development and reduces operational overhead for developers.

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Serverless Computing

What is Serverless?

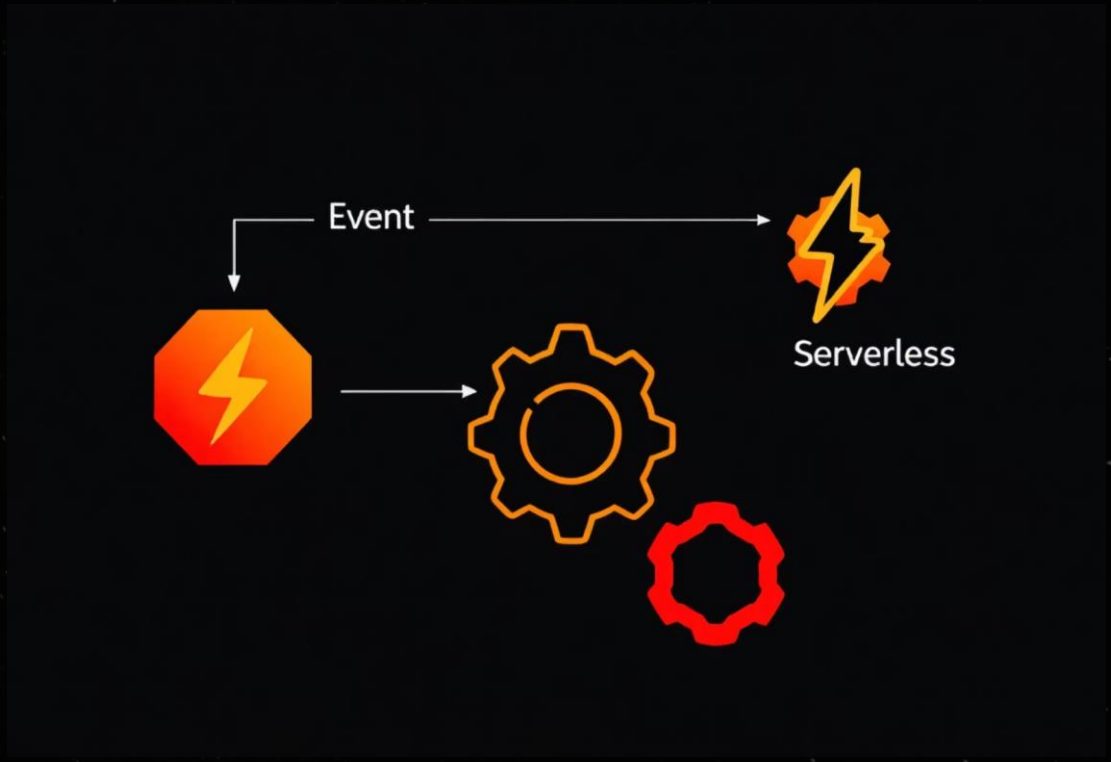
Run code without managing servers. Focus purely on logic.

Functions as a Service (FaaS)

AWS Lambda is a prime example. Code runs on demand.

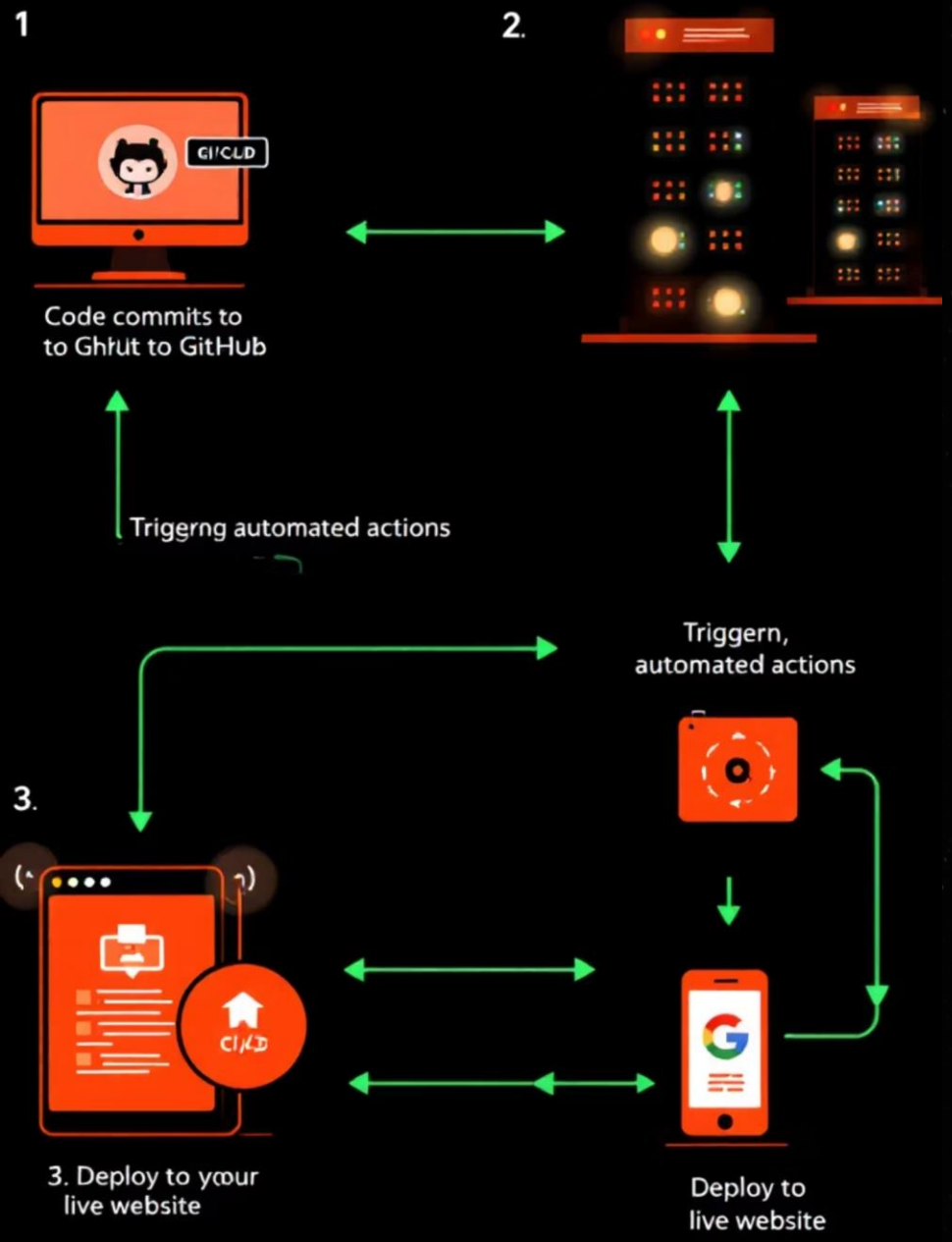
Simple Use Case

Process an image when uploaded to storage.



Serverless means you only pay for compute time used. Code scales automatically with demand. This is ideal for event-driven applications, handling bursts of traffic efficiently and cost-effectively.

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CI/CD in the Cloud

What is CI/CD?

Continuous Integration / Continuous Deployment: Automate code from development to live deployment.

Key Tools

GitHub Actions and AWS CodePipeline streamline workflows.

Deployment Automation

Changes pushed to Git automatically trigger builds and deployments. Your site goes live without manual steps.

CI/CD pipelines ensure faster, more reliable software releases. They reduce human error and facilitate rapid iteration. This practice is crucial for agile development and continuous delivery.

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Hosting a Static Website in the Cloud (Demo)

Cloud Hosting Options

AWS S3, Firebase Hosting, and Netlify are popular choices.

Core Concepts

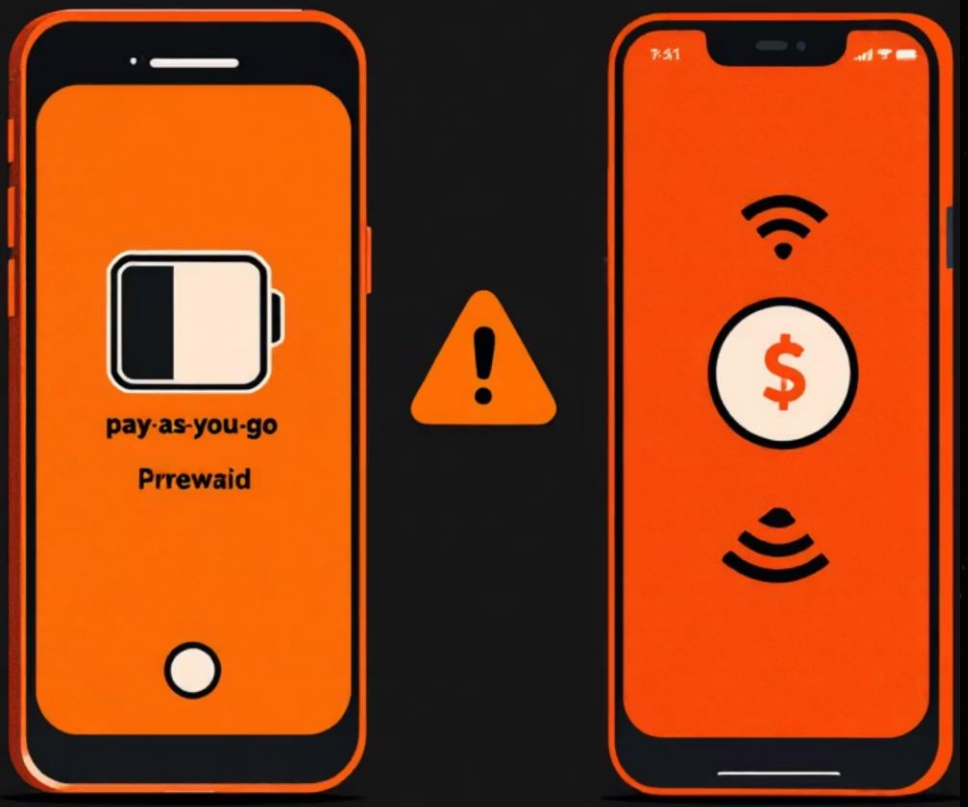
Learn about DNS, domain names, and HTTPS for security.

Simple Deployment

Upload your static files. Your portfolio site goes live globally.

Static site hosting is cost-effective and highly scalable. It's perfect for portfolios, blogs, and marketing sites. HTTPS ensures secure connections and builds trust with visitors.





Cloud Billing & Free Tier Hacks

$0

Free Tier Options

AWS, Azure, and GCP offer free tiers. Experiment risk-free.

Pricing Models

Pay-as-you-go means you only pay for what you use. Subscriptions offer predictable costs.

Avoid Surprises

Set up billing alerts. Monitor usage to prevent unexpected charges.

Understanding cloud billing is essential. Optimize resource usage. Leverage free tiers for learning and small projects. Always monitor your spending to avoid surprises.

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How Netflix Never Goes Down



Cloud Scalability

Netflix leverages cloud elasticity to handle massive global demand.

Microservices

Complex applications broken into smaller, independent services.

Auto-scaling

Automatically adjusts resources based on traffic load.

Load Balancing

Distributes incoming traffic across multiple servers efficiently.

These concepts ensure high availability and performance. They allow platforms like Netflix to deliver seamless experiences to millions of users worldwide, even during peak times.

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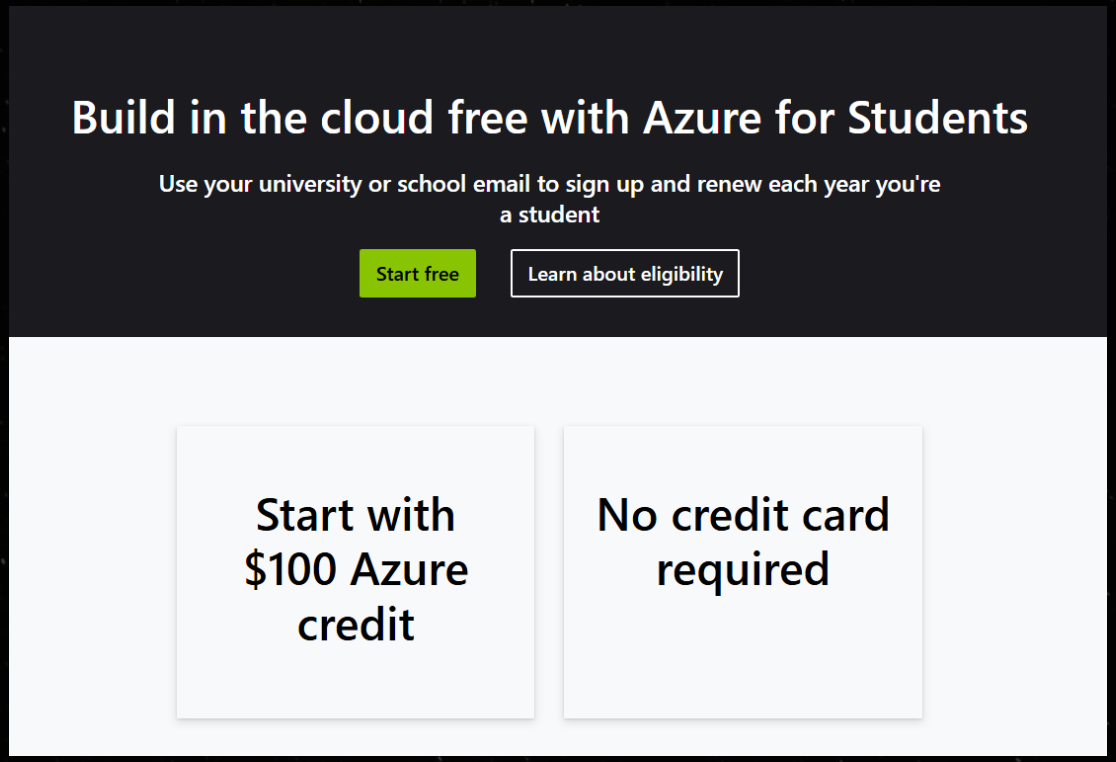
Next Steps & Resources

Access free student credits from major cloud providers.

Azure for Stude...

Explore tutorials and labs for deeper learning.

Imagine your first cloud application: What will you build?



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Build your AI startup with confidence

Microsoft offers advanced supercomputing and industry-leading security and privacy, along with access to the latest AI models and developer tools so you can build faster and stay ahead the curve.

Just getting started?

Get up to $1000 (USD) in Azure credits to build and test your idea with confidence.

$1,000 Azure credits remaining
Credits expire Sep 14, 2025

Sign up with Azure

Growing your existing company?

For startups that are a registered legal entity, apply for up to $150,000 (USD) in Azure credits and support with Microsoft.

Apply with LinkedIn

Read more about eligibility criteria here.

By submitting this application, you agree that you have read the Microsoft Privacy Statement and give permission to Microsoft for Startups to collect your data and contact you regarding your application.



Get in Touch & Start Your Tech Journey

Have questions or want to collaborate? We're here to help you grow!

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